



Logged, Not Signed: Staff Confidence After a Pool Network Retires Its Paper Sign-In Ledger

Ingeborg Nwachukwu, Renke Sabel

School of Continuous Improvement, Slop University

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Abstract. A regional network of public swimming pools has for years required both an electronic turnstile swipe and a handwritten paper-ledger entry from every visitor on arrival, the ledger nominally standing as the venue's attestable record of who was present should an incident or a dispute over the day's events arise. Across a fourteen-month window, centre managers began retiring the paper ledger at their own discretion, relying on the turnstile's existing electronic log alone, producing a natural staggered rollout: eleven centres kept the ledger throughout, thirteen dropped it. Incident-report completion rates and the median time to close a disputed incident record moved only trivially between arms, each difference sitting within the range routine month-to-month variation already produces. Staff confidence that an incident could be fully reconstructed from the centre's records fell by close to a full point on a five-point scale at ledger-dropped centres and held level where the ledger remained, a divergence a supplementary printed export of the turnstile log did not recover. Patron-facing awareness of the change was negligible throughout the window. We interpret the ledger as having functioned as a confidence-bearing ritual for the staff who kept it rather than as an operationally load-bearing record, and discuss what this suggests for institutions weighing a manual attestation step against the electronic system it quietly duplicates.

Introduction

Many venues that already log a visitor electronically also keep a manual record of the same event: a signature in a reception-desk book, a name on a clipboard, an entry in a paper diary alongside the swipe-card reader that made it redundant. Institutional accounts of organisational life describe how such formal procedures can persist as legitimating ritual well after their operational content has migrated elsewhere, retained because the appearance of a record matters independently of whether anyone consults it [1], [2]. Decoupling of this kind need not be total or sudden: a policy can remain on the books while the practice beneath it drifts for years before anyone notices or acts [3].

Whether retiring a genuinely duplicated manual record changes anything an institution actually cares about is rarely tested directly, in part because removing it is usually treated as a

foregone efficiency gain rather than an intervention worth measuring against a comparison group. Where paper and electronic versions of the same record have been compared directly, in clinical and examination settings, the electronic version is typically found at least as complete and often preferred by the staff who use it, without much attention paid to whether the underlying process the record was meant to support moves at all [4], [5], [6].

Public swimming pools offer an unusually clean setting for this question. Turnstile-swipe entry is now standard at most managed aquatic centres, yet many still keep a paper sign-in ledger at reception, ostensibly as a fallback in case the electronic log proves insufficient for reconstructing an incident or resolving a dispute. A staggered, manager-led retirement of the ledger across a single operator's network gives

us a natural quasi-experiment rather than a hypothetical one.

This paper makes three contributions, each hedged appropriately for a single-network field setting:

- We provide a field comparison, across a staggered natural rollout, of incident-report completion and dispute-resolution outcomes before and after a paper sign-in ledger's retirement, finding no operationally meaningful shift in either measure.
- We observe that the ledger's removal is associated with a decline in staff confidence that an incident could be traced through the centre's records, even though the outcomes that confidence nominally tracks do not move, suggesting the ledger's contribution may sit closer to custodial reassurance than operational function.
- We show that a low-cost attempt to recover this confidence, posting a printed export of the retained electronic log, does not do so, consistent with the possibility that the act of signing mattered more than the resulting page.

Related work

Formal structures can function as institutional myth, sustaining legitimacy independent of technical necessity and persisting even where they are loosely coupled to daily practice [1], [2]; this decoupling can hold for years without triggering correction [3]. Direct comparisons of paper and electronic versions of the same record generally find the electronic form at least as complete and, in some settings, associated with fewer missed fields or better guideline adherence, though the studies rarely test whether the process the record documents changes as a result [4], [5], [6]. Trust in an automated record is not simply a function of its accuracy: people can discount an algorithmic system's reliability sharply after even a single visible error, in a way they do not extend to an equivalent human lapse [7], though this aversion softens when users retain some ability to adjust the system themselves [8], and appropriate reliance more broadly depends on how a system's operation is understood rather than on its demonstrated performance alone [9]. Electronic monitoring introduced for one purpose can measurably reduce trust in management even where the monitored behaviour itself is unaffected [10]. A formalised

paper checklist step can materially change safety outcomes when genuinely adopted [11], but the more common finding in incident-reporting research is chronic underreporting relative to what an organisation's own paperwork implies [12]. Aquatic-facility research has concentrated on lifeguard vigilance itself, the length of an unnoticed submersion and the interventions that shorten it, rather than on the paperwork generated once an incident is noticed [13], [14], and perceptions of a resolved dispute's fairness are known to depend on the experienced process as much as the outcome [15]. Elsewhere in this University's own portfolio, an instrument built to flag an indicator for retirement did not shrink the registry it audited, with most flagged items retained under one justification or another [16]; a comparable gap between what a disposal record reports and what a physical count finds has been traced to a front-desk workaround rather than to any flaw in the record's design [17].

Method

The setting is a regional network of twenty-four publicly operated aquatic centres under one governing trust, all fitted with wristband-fob turnstiles that time-stamp every entrant electronically. Twenty-two of the twenty-four also kept a handwritten sign-in ledger at reception at the start of the study window, a legacy carryover from before the turnstile system was installed. Over the following fourteen months, individual centre managers retired the ledger at their own discretion and on their own timetable; by month fourteen, thirteen centres had dropped it and eleven had kept it, giving a staggered, manager-led quasi-experiment rather than a randomised one. Six of the thirteen ledger-dropped centres additionally began posting a printed weekly export of the turnstile log on the staff noticeboard from month eight onward, an ablation intended to test whether a paper artefact of any kind, rather than the sign-in ritual specifically, was doing the reassuring work.

Two operational outcomes were tracked across all 1,148 incidents logged network-wide over the window: incident-report completion, the share of incidents whose report was finalised to policy standard within 48 hours, and dispute-resolution time, the median days to close a formally disputed incident record. A staff confidence item, "I am confident an incident at this centre

could be fully reconstructed from our records,” was administered on a 1-5 scale to reception and duty staff at baseline and again at month fourteen; a parallel one-item patron survey asked whether visitors had noticed any change to sign-in procedure. Table 1 summarises the design.

Arm	Centres	Incidents	Staff (base-line / month 14)
Ledger re-tained	11	512	96 / 89
Ledger dropped	13	636	114 / 108
of which + printed summary	6	301	52 / 50

Table 1: Study sites, incidents logged, and staff survey response over the fourteen-month window.

The staff-confidence contrast between arms is summarised as a difference-in-differences estimate, comparing each arm’s own change from baseline to month fourteen:

$$\hat{\Delta} = (\bar{C}_{d,14} - \bar{C}_{d,0}) - (\bar{C}_{r,14} - \bar{C}_{r,0})$$

where $\bar{C}_{d,\cdot}$ and $\bar{C}_{r,\cdot}$ are the mean confidence scores for the dropped and retained arms at baseline (subscript 0) and month fourteen (subscript 14). The same estimator, applied within the dropped arm only, isolates the printed-summary ablation’s effect against the dropped-only comparison group.

Results

Incident-report completion held level across the window regardless of arm: 88.4% at ledger-retained centres against 87.6% at ledger-dropped centres, a gap well inside month-to-month variation at either group of sites (Figure 1).

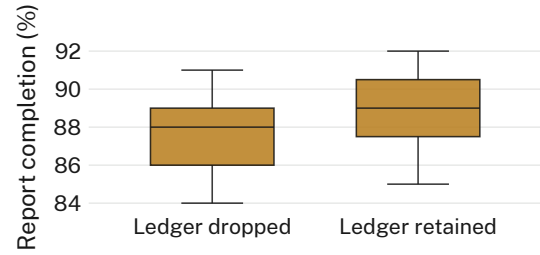


Figure 1: Per-centre incident-report completion rate over the fourteen-month window was statistically indistinguishable between arms (88.4% vs. 87.6%, n.s.).

Dispute-resolution time showed the same pattern month by month: a median of six days to close a disputed record at ledger-retained centres against six days at ledger-dropped centres, with no month in either arm departing from the network median by more than half a day (Figure 2).

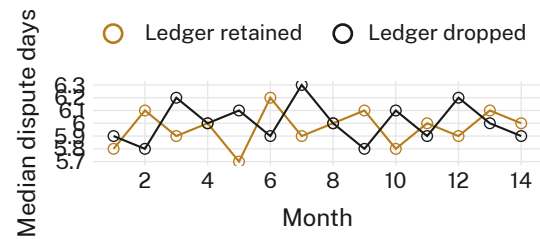


Figure 2: Monthly median dispute-resolution time held within 0.4 days of the retained-ledger baseline throughout the window, in both arms (n.s.).

Staff confidence moved in only one arm. At ledger-retained centres, confidence held from 4.1 at baseline to 4.0 at month fourteen. At ledger-dropped centres it fell from 4.0 to 3.3, a difference-in-differences estimate of $\hat{\Delta} = -0.8$ points ($p < .05$). The six centres that additionally posted a printed weekly export of the turnstile log from month eight showed no recovery: 3.4 at month fourteen against 3.3 at the other seven dropped centres, an ablation difference of 0.1 points (n.s.); the printed export is, in any case, still posted at all six centres as of writing (Figure 3).

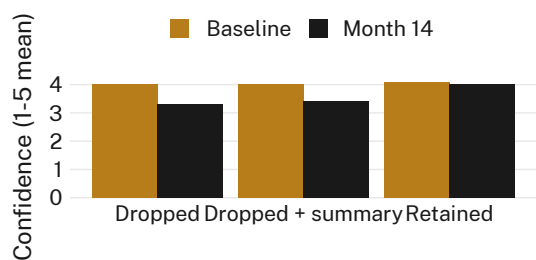


Figure 3: Staff confidence in incident traceability fell only where the paper ledger was retired (4.0 to 3.3, $p < .05$); a printed weekly export of the turnstile log did not recover it (3.3 vs. 3.4, n.s.).

The parallel patron-awareness item found the change essentially invisible from the visitor side: fewer than 4% of surveyed patrons at ledger-dropped centres correctly identified that a sign-in step had been removed, a proportion indistinguishable from the false-positive rate at ledger-retained centres where no change had occurred.

Discussion

The pattern across all three measures is consistent with the ledger's function sitting closer to staff reassurance than to operational necessity. Neither incident-report completion nor dispute-resolution time moved when the ledger was withdrawn, which is what we would expect if the turnstile log already carried the operationally relevant information and the ledger's entries were rarely, if ever, the record actually consulted. Staff confidence moving regardless suggests the ledger's contribution was not the content of the page but the fact of the signing: a visible, effortful act that the electronic swipe, however complete, does not visibly perform in front of the person relying on it. The printed-summary ablation's failure to recover confidence is consistent with this reading, since it supplies the same underlying data as the ledger without the act of attesting to it. We observe this pattern most clearly among reception and duty staff, who complete the confidence item with the ledger's absence freshly in view; we would not be surprised if the effect proves smaller, or slower to appear, among staff with less day-to-day contact with the sign-in point itself.

Limitations

Our sample is drawn from one regional network and one facility type, though the consistency of the null finding on operational outcomes across

eleven retained and thirteen dropped centres suggests the pattern is not an artefact of any single site's practice. The printed-summary ablation covers only six centres, which limits its power to detect a genuinely small recovery effect; the point estimate, however, sits closer to zero than to the confidence gap it was intended to close. Centre managers, not a randomiser, chose when to retire the ledger, and it remains possible that centres already trending toward lower staff confidence were also more inclined to drop it, though no baseline difference between eventual arms was detected before the rollout began.

Conclusion

Retiring a paper sign-in ledger already duplicated by an electronic turnstile log left incident-report completion and dispute-resolution time unmoved, but measurably reduced staff confidence that an incident could be traced through the centre's own records, a gap a printed summary of the very data the ledger duplicated did not close. Future work might test whether the confidence effect fades with time since retirement, or whether it can be recovered by a staff-facing attestation step that keeps the act of signing without reinstating the page.

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